MS-ADS Program - University of Chicago
Statistical Models for Data Science Autumn 2025 - **Group Assignment 5** (Submission: 12/11/2025)
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## 1.0 Overview

We used five different statistical models of the daily count of serious crashes, with predictors: Day of week, Month of year, Proportion of crashes that day with bad roads (not DRY or UNKNOWN), Proportion of crashes that day with bad weather (not CLEAR or UNKNOWN).

The best-fitting model was a negative binomial regression with a log link (we will refer to this as “the day-level model” in this report). All of the results below come from that model.

## 1.1. How weekday and month affect serious crashes

## 1.2 Weekday patterns

From the model, the average number of serious crashes per day by weekday is: Monday: 5.4, Tuesday: 5.39, Wednesday: 5.34, Thursday: 5.37, Friday: 6.0, Saturday: 6.66, Sunday: 5.91

The model’s fitted values line up almost exactly with these observed averages. The key takeaways are:

1. Weekends are riskier on a per-day basis. Saturdays see about 1.3 more serious crashes per day than Wednesdays on average.

2. Fridays and Sundays are also elevated. Fridays average about 6 serious crashes/day, Sundays about 5.9, compared with midweek around 5.3 - 5.4.

3. These differences remain after accounting for months, bad roads, and bad weather in the model.

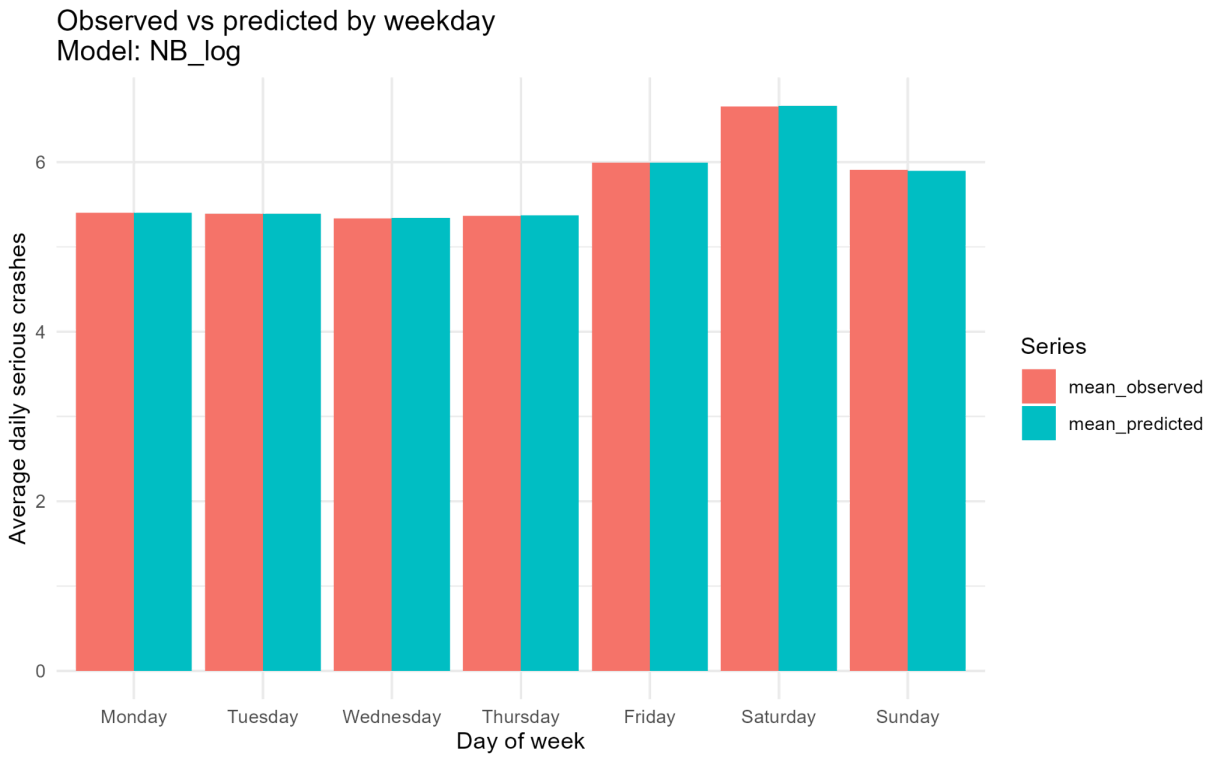
## 1.3 Possible real-world explanations:

1. More discretionary travel on weekends; more exposure, often later at night.

2. Higher rates of alcohol/drug involvement or speeding on weekend nights, which aligns with national patterns.

3. More pedestrians and cyclists in some areas on weekends, different driving patterns.

**Fig 2:**




### 1.4 Monthly/seasonal patterns

Average daily serious crashes by month are:

- **Winter (Jan–Mar):** Jan: 4.46, Feb: 4.68, Mar: 4.93
- **Spring (Apr–May):** Apr: 4.90, May: 6.17
- **Summer (Jun–Aug):** Jun: 6.79, Jul: 7.14, Aug: 6.52
- **Fall (Sep–Nov):** Sep: 6.43, Oct: 6.27, Nov: 5.20
- **Dec:** 5.10

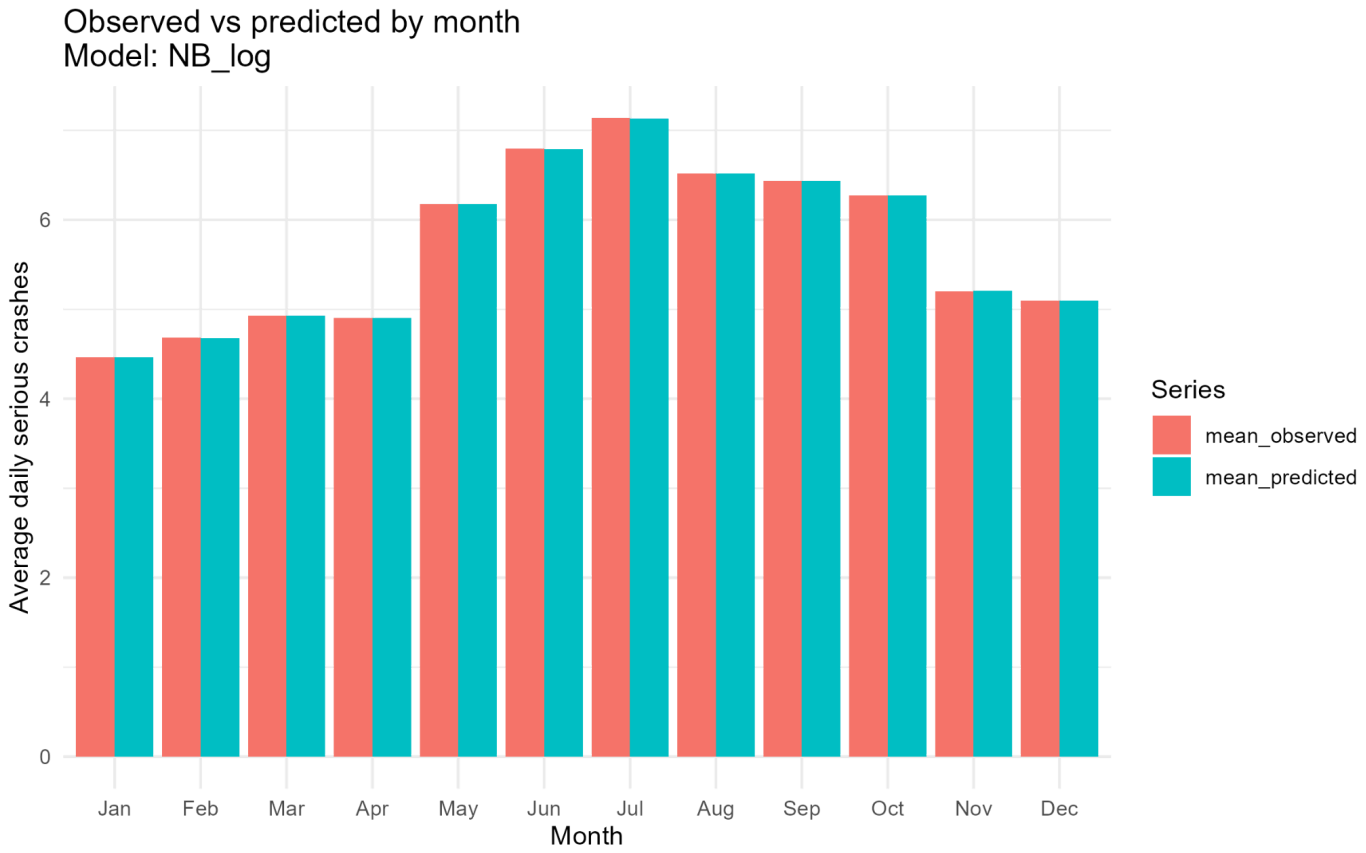
Again, the model's predictions are basically identical to these averages. Key takeaways are:

1. Strong seasonal pattern. Serious crashes are lowest in winter and highest in mid-summer, peaking at about 7.1 serious crashes per day in July.
2. Relative to January, July has roughly 2.7 extra serious crashes per day, which is about a 60% increase in the daily serious crash count.
3. The increase starts around May, stays elevated through September, and then declines into late fall and winter.

**1.5 Possible real-world explanations:**

1. More people moving around in good weather, increasing exposure.
2. More evening and night-time activity in warm months.
3. More vulnerable road users are active in summer, increasing opportunities for severe conflicts even if per-exposure risk is similar.

**Fig. 2**:



### 2.0. Road surface and weather conditions

The model includes:

- **prop_bad_road:** share of that day's crashes on non-dry (or non-unknown) surfaces
- **prop_bad_weather:** share of that day's crashes in non-clear (or non-unknown) weather

In the best model, the bad road effect is very small and statistically insignificant (coefficient ≈ 0.008, p ≈ 0.96). The bad weather effect is modestly negative and not statistically significant (coefficient ≈ - 0.24, p ≈ 0.22). To put this into "real day" terms, we looked at predicted serious crashes on a Monday in January, for combinations of bad road and bad weather. From the `road_weather_grid`, bad weather changes (holding bad roads fixed at median):

- Low bad weather (≈0.3% of crashes): 4.42 serious crashes/day
- Typical bad weather day (≈5% of crashes): 4.42
- High bad weather day (≈48% of crashes): 3.99

So, going from a "mostly clear" day to a "half of all crashes in bad weather" day lowers the predicted serious crash count slightly (by 0.4 - 0.5 crashes per day), but the effect is small compared to weekday/month differences. It is not statistically robust in this model. Bad roads vary more in the data, but even across that range, we predicted serious crashes move only from about 4.47 to 4.50 (with low bad weather), or 3.99 to 4.01 (with high bad weather).

**Interpretation/hypothesis:**

1. At the daily, citywide level, once we account for how many trips and what kinds of trips people are making (captured indirectly by month and weekday), changes in the share of crashes occurring in "bad weather" or on "bad roads" do not strongly change the number of serious crashes that day.
2. One plausible behavioral story: on truly bad weather days, total traffic volumes drop, and drivers may slow down and drive more cautiously, offsetting some of the added risk per trip. So we might see more minor crashes per trip, but not a large increase in serious crashes per day.
3. These results do not say that bad weather or poor road surfaces are "safe"; they do suggest that at the day-level, citywide, those conditions are not the main drivers of how many serious crashes we see.

For CDOT's purposes, this supports focusing more on exposure patterns (time of week, time of year, land use, trip purposes) than on expecting daily bad-weather variation to explain big swings in serious crash counts.

### 3.0. Model performance and when to trust it

**3.1 Overall fit:** We compared five models.

**Table 1:**

| Model | AIC | RMSE |
|---|---|---|
| Negative binomial (log) | 8551.7 | 2.55 |
| ZINB full (all preds in count, intercept-only zero) | 8554 | 2.55 |
| Poisson (log) | 8565 | 2.55 |
| Poisson (identity) | 8572 | 2.56 |
| ZINB split | 8589 | 2.56 |

The negative binomial (log link) model has the lowest AIC, meaning it strikes the best balance between fit and complexity. The RMSE (typical error in daily serious crash counts) is about 2.55 crashes per day, and this is almost identical across the models. That tells us the core signal is captured similarly, but NB does it with a more appropriate distribution for over dispersed count data. The dispersion parameter (theta ≈ 42) shows mild over-dispersion, enough that NB improves fit vs pure Poisson, but not so large that we need a very complex model.

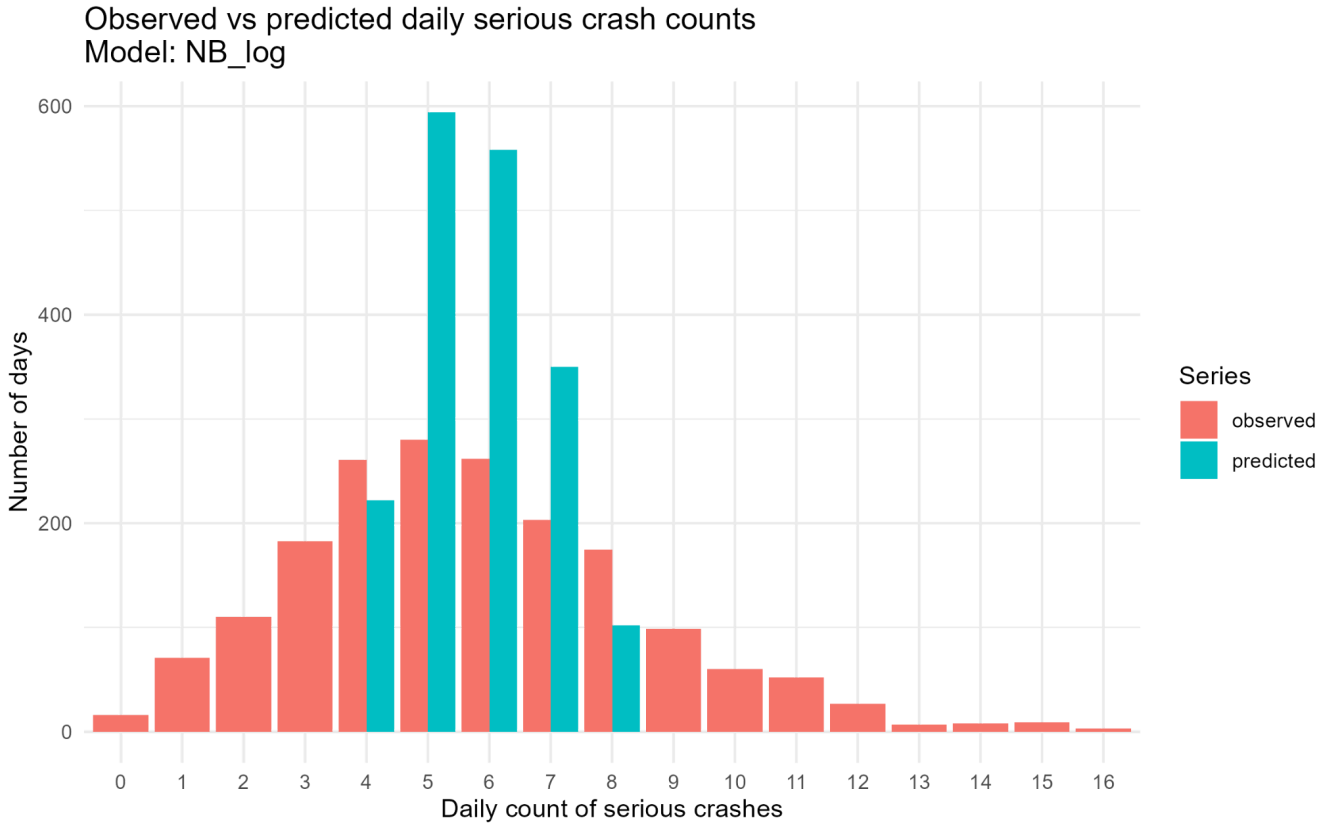
### 3.2 Predicted and Observed

**Method 1:**

This method transforms the model's continuous expected values into discrete counts to compare against observed data. For every observation (each day), the model calculates the predicted mean based on the covariates (Road condition, Weather Condition, Day of Week, Month).

The observed vs predicted frequency distribution of daily serious crash counts shows that the model reproduces the shape of the distribution, i.e. how many days have 3, 4, or 5 serious crashes. There is still some spread which is a typical error of ±2-3 crashes on any given day. Please refer to figure 3.

**Fig 3:**

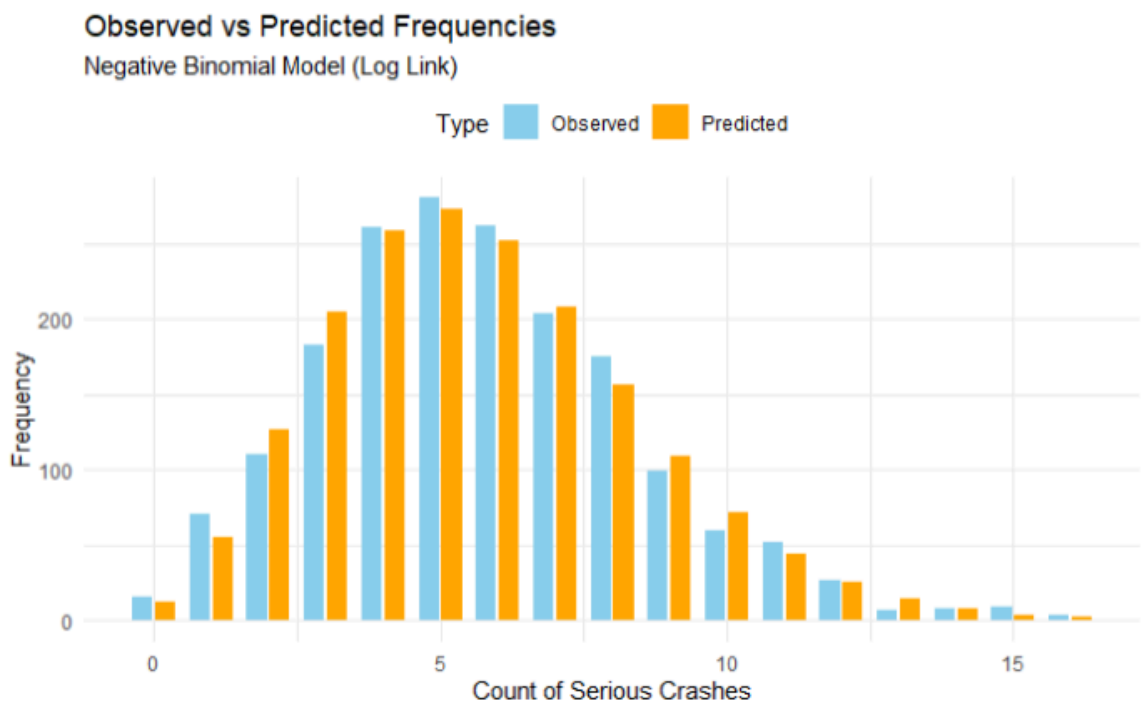



**Method 2:**

This method uses the full probability mass function (PMF) of the estimated Negative Binomial distribution to calculate the expected frequency of each count level across the dataset. It utilizes both the predicted mean for each day and the model's global dispersion parameter to define a unique probability distribution for every specific day.

The model accurately captures the frequency of safe days (e.g., predicting ~12 zero-crash days vs. 16 actual) demonstrating that it does not overlook low-risk periods. Likewise, it correctly identifies the frequency of dangerous days (e.g., predicting ~8 days with 14 crashes vs. 8 actual) proving that it effectively captures the high-risk tail. Furthermore, it precisely mirrors the volume of typical days (counts 4–6), matching the observed central peak without over-concentrating predictions at the mean.

**Fig 4:**




**3.3 When this model is useful**:

A. **Understanding broad patterns**: how serious crashes vary by weekday and month. Also, whether “bad road” and “bad weather” days are systematically different at the daily level.

B. **Planning and prioritization**: identifying that weekends and summer months are periods of elevated serious crash risk, even after adjusting for weather and road conditions. Additionally, informing enforcement, education, and operations scheduling.

**3.4 Not good uses:**

A. Random events, special circumstances, and local spikes will still cause day-to-day variation that this model cannot foresee.

B. This is a citywide day-level model; it does not indicate which specific corridors or intersections are the most dangerous. For that, CDOT should continue using detailed spatial and crash-level models.

C. The model treats all serious crashes as one aggregate; disaggregated modeling would be needed for group-specific behavior.

In short, the model is most useful as a context and confounding check for ongoing crash-level Vision Zero analyses. It suggests that a substantial part of the variation in serious crashes is systematic by weekday and season. While daily changes in bad weather/road conditions do not appear to be the main drivers of citywide serious crash counts, once exposure is accounted for.

## Appendix

### A1. Methodology

Daily serious crash counts were modeled using generalized linear models for counts. First, crash-level data from the City of Chicago were aggregated to the day level, with the response defined as the daily count of serious crashes (at least one fatal or incapacitating injury). Predictors were: the proportion of crashes with bad road conditions (ROADWAY_SURFACE_COND not DRY/UNKNOWN), the proportion of crashes with bad weather (WEATHER_CONDITION not CLEAR/UNKNOWN), weekday, and month.

Five models were fit: Poisson with identity link, Poisson with log link, negative binomial with log link (glm.nb), and two zero-inflated negative binomial (ZINB) models. The ZINB models were implemented via custom maximum likelihood, using optim() to maximize the log-likelihood of a mixture of a point mass at zero and a negative binomial count component. Model performance was evaluated using Akaike Information Criterion (AIC) and root mean squared error (RMSE) based on fitted values on the original count scale. The negative binomial log-link model had the best AIC and was therefore used for interpretation.

### A2. Model comparison

From the `model_comp` output

**Table 2:**

| Model | AIC | RMSE |
|---|---|---|
| Negative binomial (log link) - `NB_log` | 8551.7 | 2.55 |
| ZINB full - all predictors in count, intercept-only zero | 8554 | 2.55 |
| Poisson (log link) - `Poisson_log` | 8565 | 2.55 |
| Poisson (identity link) - `Poisson_identity` | 8572 | 2.56 |
| ZINB split - weekday/month in count, bad road/weather in zero | 8589 | 2.56 |

### A3. Rationale for selecting the final model

1. The negative binomial with log link has the lowest AIC (8551.7), indicating the best trade-off between goodness-of-fit and model complexity.

2. All models have very similar RMSE (~2.55 - 2.56), so mean prediction error alone wouldn't distinguish them. Given that, AIC is the more appropriate criterion since it incorporates the likelihood and penalizes extra parameters.

3. The outcome is a count of serious crashes per day with mild overdispersion relative to Poisson. The negative binomial family is standard for this situation and more flexible than Poisson while still fairly simple.

4. The two zero-inflated negative binomial models (`ZINB_full` and `ZINB_split`) do not improve AIC relative to `NB_log`.

5. `NB_log` is comparatively easy to interpret due to multiplicative effects on the expected daily count. For CDOT staff, this makes it easier to translate model output into “X% higher on Saturdays than Mondays,”.

For these reasons, we selected the negative binomial (log link) model as the final model for all substantive interpretation in the main report.

**A4. R code**

```
library(tidyverse)
library(lubridate)
library(MASS)   # for glm.nb (negative binomial)
library(ggplot2)
fit_zinb <- function(formula_count, formula_zero, data, maxit = 1000) {
  # Model frame and matrices for count part
  mf_count <- model.frame(formula_count, data = data)
  y <- model.response(mf_count)
  X <- model.matrix(attr(mf_count, "terms"), mf_count)
  mf_zero <- model.frame(formula_zero, data = data)
  Z <- model.matrix(attr(mf_zero, "terms"), mf_zero)
  n <- length(y)
  p <- ncol(X)
  q <- ncol(Z)
 pois_fit <- glm(formula_count, family = poisson(link = "log"), data = data)
  beta_start <- coef(pois_fit)
  if (length(beta_start) != p) {
            # in case of dropped columns, pad with zeros
            beta_start_full <- rep(0, p)
            names(beta_start_full) <- colnames(X)
            beta_start_full[names(beta_start)] <- beta_start
            beta_start <- beta_start_full
  }
  gamma_start <- rep(0, q)  # start zero-inflation effects at 0
  log_theta_start <- 0              # theta = exp(0) = 1

  par_start <- c(beta_start, gamma_start, log_theta_start)
  negloglik <- function(par) {
            beta <- par[1:p]
            gamma <- par[(p + 1):(p + q)]
            log_theta <- par[p + q + 1]
            theta <- exp(log_theta)
            eta <- as.vector(X %*% beta)
            mu  <- exp(eta)  # NB mean
            zeta <- as.vector(Z %*% gamma)
            pi   <- 1 / (1 + exp(-zeta))  # logistic
            loglik_vec <- numeric(n)
            for (i in seq_len(n)) {
            if (y[i] == 0) {
            # P(Y=0) = pi + (1 - pi)*NB(0)
            p_nb0 <- dnbinom(0, mu = mu[i], size = theta)
            prob  <- pi[i] + (1 - pi[i]) * p_nb0
            } else {
            # P(Y=y) = (1 - pi) * NB(y)
            p_nby <- dnbinom(y[i], mu = mu[i], size = theta)
            prob  <- (1 - pi[i]) * p_nby
            }
            prob <- max(prob, 1e-16) # avoid log(0)
            loglik_vec[i] <- log(prob)
            }
            -sum(loglik_vec)
  }
  opt <- optim(
            par = par_start,
            fn = negloglik,
            method = "BFGS",
            control = list(maxit = maxit)
  )
```

```
  par_hat <- opt$par
  beta_hat <- par_hat[1:p]
  gamma_hat <- par_hat[(p + 1):(p + q)]
  log_theta_hat <- par_hat[p + q + 1]
  theta_hat <- exp(log_theta_hat)

  beta <- beta_hat
  gamma <- gamma_hat
  eta <- as.vector(X %*% beta)
  mu  <- exp(eta)
  zeta <- as.vector(Z %*% gamma)
  pi   <- 1 / (1 + exp(-zeta))

  fitted_mean <- (1 - pi) * mu

  logLik_val <- -opt$value
  k <- length(par_hat)
  AIC_val <- -2 * logLik_val + 2 * k

  list(
            coefficients = list(
            count = beta_hat,
            zero  = gamma_hat
            ),
            log_theta = log_theta_hat,
            logLik = logLik_val,
            AIC = AIC_val,
            df = k,
            fitted = fitted_mean,
            converged = (opt$convergence == 0),
            formula_count = formula_count,
            formula_zero  = formula_zero
  )
}

crashes_raw <- readr::read_csv("crashes.csv")
crashes <- crashes_raw %>%
  mutate(
            crash_date = as.Date(CRASH_DATE),
            serious = INJURIES_FATAL > 0 | INJURIES_INCAPACITATING > 0,
            bad_weather = !(WEATHER_CONDITION %in% c("CLEAR", "UNKNOWN")),
            bad_road    = !(ROADWAY_SURFACE_COND %in% c("DRY", "UNKNOWN"))
  )

daily_agg <- crashes %>%
  group_by(crash_date) %>%
  summarise(
            n_crashes              = n(),
            serious_count          = sum(serious, na.rm = TRUE),
            prop_bad_weather = mean(bad_weather, na.rm = TRUE),
            prop_bad_road        = mean(bad_road, na.rm = TRUE),
            .groups = "drop"
  )

all_days <- tibble(
  crash_date = seq(min(daily_agg$crash_date),
            max(daily_agg$crash_date),
            by = "day")
)

daily <- all_days %>%
  left_join(daily_agg, by = "crash_date") %>%
  mutate(
            n_crashes              = if_else(is.na(n_crashes), 0L, n_crashes),
            serious_count          = if_else(is.na(serious_count), 0L, serious_count),
            prop_bad_weather = if_else(is.na(prop_bad_weather), 0, prop_bad_weather),
            prop_bad_road        = if_else(is.na(prop_bad_road), 0, prop_bad_road),

            weekday = factor(
            weekdays(crash_date),
            levels = c("Monday","Tuesday","Wednesday",
             "Thursday","Friday","Saturday","Sunday")
            ),
            month   = factor(month(crash_date, label = TRUE, abbr = TRUE))
  )
```

```
cat("Number of days in daily dataset:", nrow(daily), "\n\n")

form_full <- serious_count ~ prop_bad_road + prop_bad_weather + weekday + month
form_zinb_count1 <- form_full
form_zinb_zero1  <- ~ 1

form_zinb_count2 <- serious_count ~ weekday + month
form_zinb_zero2  <- ~ prop_bad_road + prop_bad_weather

m_pois_identity <- glm(
  form_full,
  family = poisson(link = "identity"),
  data   = daily
)

m_pois_log <- glm(
  form_full,
  family = poisson(link = "log"),
  data   = daily
)

m_nb_log <- glm.nb(
  form_full,
  data = daily
)

m_zinb_full <- fit_zinb(
  formula_count = form_zinb_count1,
  formula_zero  = form_zinb_zero1,
  data = daily
)

m_zinb_split <- fit_zinb(
  formula_count = form_zinb_count2,
  formula_zero  = form_zinb_zero2,
  data = daily
)

rmse <- function(obs, pred) sqrt(mean((obs - pred)^2))
obs <- daily$serious_count
pred_pois_identity <- fitted(m_pois_identity)
pred_pois_log      <- fitted(m_pois_log)
pred_nb_log        <- fitted(m_nb_log)
pred_zinb_full     <- m_zinb_full$fitted
pred_zinb_split    <- m_zinb_split$fitted

model_comp <- tibble(
  model = c("Poisson_identity",
            "Poisson_log",
            "NB_log",
            "ZINB_full",
            "ZINB_split"),
  AIC = c(
            AIC(m_pois_identity),
            AIC(m_pois_log),
            AIC(m_nb_log),
            m_zinb_full$AIC,
            m_zinb_split$AIC
  ),
  RMSE = c(
            rmse(obs, pred_pois_identity),
            rmse(obs, pred_pois_log),
            rmse(obs, pred_nb_log),
            rmse(obs, pred_zinb_full),
            rmse(obs, pred_zinb_split)
  )
) %>%
  arrange(AIC, RMSE)

cat("=== Model comparison (for appendix) ===\n")
print(model_comp)
cat("\n")

best_model_name <- model_comp$model[1]
```

```
cat("Best model by AIC (tie-broken by RMSE):", best_model_name, "\n\n")

model_list <- list(
  Poisson_identity = m_pois_identity,
  Poisson_log      = m_pois_log,
  NB_log           = m_nb_log,
  ZINB_full        = m_zinb_full,
  ZINB_split       = m_zinb_split
)

pred_list <- list(
  Poisson_identity = pred_pois_identity,
  Poisson_log      = pred_pois_log,
  NB_log           = pred_nb_log,
  ZINB_full        = pred_zinb_full,
  ZINB_split       = pred_zinb_split
)

best_model <- model_list[[best_model_name]]
daily$pred_serious <- pred_list[[best_model_name]]

freq_df <- daily %>%
  transmute(
        observed  = serious_count,
        predicted = round(pred_serious)
  ) %>%
  pivot_longer(
        cols = c(observed, predicted),
        names_to = "type",
        values_to = "count"
  ) %>%
  count(type, count, name = "frequency") %>%
  arrange(type, count)

cat("=== Frequency of observed vs predicted daily serious crashes ===\n")
print(freq_df)
cat("\n")

p1 <- ggplot(freq_df, aes(x = factor(count), y = frequency, fill = type)) +
  geom_col(position = "dodge") +
  labs(
        x = "Daily count of serious crashes",
        y = "Number of days",
        fill = "Series",
        title = paste("Observed vs predicted daily serious crash counts\nModel:", best_model_name)
  ) +
  theme_minimal()

print(p1)
ggsave("freq_obs_pred.png", p1, width = 8, height = 5, dpi = 300)

weekday_effects <- daily %>%
  group_by(weekday) %>%
  summarise(
        mean_observed  = mean(serious_count),
        mean_predicted = mean(pred_serious),
        .groups = "drop"
  )

cat("=== Weekday effects (observed vs predicted means) ===\n")
print(weekday_effects)
cat("\n")

p2 <- ggplot(
  weekday_effects %>%
        pivot_longer(c(mean_observed, mean_predicted),
        names_to = "type", values_to = "value"),
  aes(x = weekday, y = value, fill = type)
) +
  geom_col(position = "dodge") +
  labs(
        x = "Day of week",
        y = "Average daily serious crashes",
        fill = "Series",
        title = paste("Observed vs predicted by weekday\nModel:", best_model_name)
```

```
  ) +
  theme_minimal()

print(p2)
ggsave("weekday_effects.png", p2, width = 8, height = 5, dpi = 300)

month_effects <- daily %>%
  group_by(month) %>%
  summarise(
            mean_observed  = mean(serious_count),
            mean_predicted = mean(pred_serious),
            .groups = "drop"
  )

cat("=== Month effects (observed vs predicted means) ===\n")
print(month_effects)
cat("\n")

p3 <- ggplot(
  month_effects %>%
            pivot_longer(c(mean_observed, mean_predicted),
            names_to = "type", values_to = "value"),
  aes(x = month, y = value, fill = type)
) +
  geom_col(position = "dodge") +
  labs(
            x = "Month",
            y = "Average daily serious crashes",
            fill = "Series",
            title = paste("Observed vs predicted by month\nModel:", best_model_name)
  ) +
  theme_minimal()

print(p3)
ggsave("month_effects.png", p3, width = 8, height = 5, dpi = 300)

cat("=== Best model summary (custom) ===\n")
if (best_model_name %in% c("Poisson_identity", "Poisson_log", "NB_log")) {
  print(summary(best_model))
} else {
  cat("Zero-inflated NB model selected.\n")
  cat("Count (NB mean) coefficients:\n")
  print(best_model$coefficients$count)
  cat("\nZero-inflation (logit) coefficients:\n")
  print(best_model$coefficients$zero)
  cat("\nlog(theta):", best_model$log_theta, "\n")
  cat("logLik:", best_model$logLik, " AIC:", best_model$AIC, "\n")
}
cat("\n")

ref_weekday <- levels(daily$weekday)[1]
ref_month   <- levels(daily$month)[1]

road_quantiles  <- quantile(daily$prop_bad_road,   probs = c(0.1, 0.5, 0.9), na.rm = TRUE)
weath_quantiles <- quantile(daily$prop_bad_weather, probs = c(0.1, 0.5, 0.9), na.rm = TRUE)

road_weather_grid <- expand_grid(
  prop_bad_road      = road_quantiles,
  prop_bad_weather = weath_quantiles,
  weekday = ref_weekday,
  month   = ref_month
)

if (best_model_name %in% c("Poisson_identity", "Poisson_log", "NB_log")) {
  road_weather_grid$pred_serious <- predict(
            best_model,
            newdata = road_weather_grid,
            type = "response"
  )
} else if (best_model_name == "ZINB_full") {
  mf_count_new <- model.frame(form_zinb_count1, data = road_weather_grid)
  X_new <- model.matrix(attr(model.frame(form_zinb_count1, daily), "terms"), mf_count_new)
  mf_zero_new <- model.frame(form_zinb_zero1, data = road_weather_grid)
  Z_new <- model.matrix(attr(model.frame(form_zinb_zero1, daily), "terms"), mf_zero_new)
```

```
  beta <- best_model$coefficients$count
  gamma <- best_model$coefficients$zero

  eta_new <- as.vector(X_new %*% beta)
  mu_new  <- exp(eta_new)
  zeta_new <- as.vector(Z_new %*% gamma)
  pi_new   <- 1 / (1 + exp(-zeta_new))

  road_weather_grid$pred_serious <- (1 - pi_new) * mu_new

} else if (best_model_name == "ZINB_split") {
  mf_count_new <- model.frame(form_zinb_count2, data = road_weather_grid)
  X_new <- model.matrix(attr(model.frame(form_zinb_count2, daily), "terms"), mf_count_new)
  mf_zero_new <- model.frame(form_zinb_zero2, data = road_weather_grid)
  Z_new <- model.matrix(attr(model.frame(form_zinb_zero2, daily), "terms"), mf_zero_new)

  beta <- best_model$coefficients$count
  gamma <- best_model$coefficients$zero

  eta_new <- as.vector(X_new %*% beta)
  mu_new  <- exp(eta_new)
  zeta_new <- as.vector(Z_new %*% gamma)
  pi_new   <- 1 / (1 + exp(-zeta_new))

  road_weather_grid$pred_serious <- (1 - pi_new) * mu_new
}

cat("=== Predicted serious crashes at combinations of bad road / bad weather ===\n")
print(road_weather_grid)
cat("\n")
```